EVQ-Cosh: Optimal Frequency Allocation for Rotary Position Embedding

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Abstract

Rotary position embedding (RoPE) is governed by three design axes: the base frequency, the head dimension, and the allocation of frequencies across channels. The first two have received significant attention; the third—how frequency mass is distributed—has remained fixed at the geometric (equal-ratio) schedule since RoFormer, despite being a key determinant of positional representational quality. We formulate frequency allocation as a variational inverse problem, derive a broadband surrogate whose stationary density satisfies a closed-form ODE, and obtain a one-parameter family (EVQ-cosh) in which geometric RoPE is the degenerate $\tau{=}0$ limit. EVQ is architecture-agnostic: it modifies only the inverse-frequency initialization, adds zero parameters, and composes with existing inference-time methods. On standard multi-head attention (MHA), EVQ+YaRN reaches 100% retrieval at 8K (3 seeds per configuration, 6 total) where Geo+YaRN plateaus at 61%, and lowers PPL by 14–32% at 8–16K. On Multi-head Latent Attention (MLA, DeepSeek-V2/V3), where only 16 frequency channels carry positional information, EVQ alone outperforms Geo+YaRN at $2\times$ extrapolation (-31.1% PPL, 3 seeds). In PE-dominant extreme extrapolation, EVQ surpasses learnable positional encoding with zero extra parameters (333.7 vs. 455.3 PPL). These results identify frequency allocation as a fundamental and under-explored design axis for rotary position embedding, with consistent gains across MHA, MLA, and video DiT architectures.

1 Introduction

The quality of rotary position embedding (RoPE) is governed by three design axes: the base frequency b, the head dimension d_{head}, and the allocation of frequencies across channels. The first axis has received sustained attention—NTK-aware scaling, YaRN, and related methods all operate by adjusting b or its effective value at inference time [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024]. The second axis is typically set by architecture constraints and has become newly relevant as compressed designs such as Multi-head Latent Attention (MLA) reduce the number of RoPE channels from 64 to as few as 16. The third axis—how frequency mass is distributed among channels—has remained essentially unchanged since RoFormer [Su et al., 2024]: every mainstream implementation uses a geometric (equal-ratio) schedule.

This paper shows that geometric allocation is a degenerate limiting case, not a generic optimum. We formulate frequency allocation as a variational inverse problem whose stationary density satisfies a closed-form ODE, yielding a one-parameter family (EVQ-cosh) where geometric RoPE is the $\tau{=}0$ limit. The method modifies only the inverse-frequency initialization, adds zero parameters, and is a drop-in replacement for any RoPE-based model. Three empirical findings span different architectures:

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MHA: EVQ unlocks inference-time scaling. On standard multi-head attention (50M–750M), EVQ provides a substantially better training-time substrate for inference-time methods. Across three seeds per configuration (six total), EVQ+YaRN reaches 100% retrieval at 8K while Geo+YaRN remains at 61%, with PPL improvements of 14–32% at 8–16K. Progressive context extension (single seed) widens this gap monotonically, extending functional context to 48K ($24\times$ training length) at PPL 2.63.

MLA: principled allocation matters more when channels are scarce. On MLA (432M, 3-seed), where only 16 frequency channels carry positional information, EVQ alone reduces $2\times$ extrapolation PPL by 31.1% and outperforms Geo+YaRN at 16K. When every channel's placement counts, the gap between geometric and optimal allocation is amplified.

PE-dominant regime: frequency structure alone is decisive. In DAPE-style extreme extrapolation ($128\rightarrow8$K), EVQ surpasses learnable adaptive positional encoding with zero additional parameters (333.7 vs. 455.3 PPL), isolating frequency allocation itself as the decisive variable.

The central claim is that frequency allocation is a fundamental, under-explored PE design axis. Geometric RoPE has two orthogonal deficiencies: a *shape* deficiency (uniform density wastes channels) and a *range* deficiency (frequencies designed for L_{train} cannot cover longer distances). EVQ addresses shape; YaRN addresses range; their benefits compose. Our contributions:

- We identify frequency allocation as a fundamental PE design axis and derive EVQ-cosh, a closed-form variational solution in which geometric RoPE is the $\tau = 0$ degenerate case of a broader optimal family.
- We demonstrate architecture-agnostic gains: EVQ improves MHA (50M–750M text), MLA (432M, 16 RoPE channels), and video DiT. A single semi-analytic formula $\tau^* = d_{\text{head}}/\sqrt{L}$—sub-optimal by construction—nonetheless selects effective allocations across all tested settings ($< 1\%$ PPL gap from per-setting optima).
- We show that the strongest systems result is the composition: EVQ unlocks YaRN, with training-time and inference-time positional optimization being orthogonal and strongly complementary.
- We show that progressive training widens EVQ's advantage monotonically (single seed): three-stage context extension ($512\rightarrow1024\rightarrow2048$) yields a growing relative PPL advantage ($-34.6\%\rightarrow-52.0\%\rightarrow-81.2\%$ at 16K), extending functional context to 48K ($24\times$ training length) at PPL 2.63.

2 Related Work

Positional encoding design space. Transformers rely on positional structure to recover order information [Vaswani et al., 2017]. The design space includes relative attention biases [Shaw et al., 2018, Raffel et al., 2020], ALiBi [Press et al., 2022], and rotary position embedding (RoPE) [Su et al., 2024]. Within RoPE, most prior work focuses on the base frequency axis (NTK-aware scaling, YaRN) or inference-time rescaling. Our work targets a different axis: the allocation of frequencies across channels, which has remained at the geometric default since RoFormer.

Inference-time context extension. Recent long-context methods preserve the training checkpoint and modify positional behavior only at inference time [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024, Zhu et al., 2024, Jin et al., 2024]. LongRoPE2 [Shang et al., 2025] searches for per-channel scaling factors via evolution, optimizing inference-time rescaling on a frozen checkpoint. CLEX [Chen et al., 2024] trains a continuous length-extrapolation module. While both LongRoPE2 and EVQ operate on per-channel frequency structure, they target orthogonal design axes: EVQ modifies the frequency layout *before training*, whereas LongRoPE2 searches for rescaling *after training*. This complementarity is demonstrated empirically: EVQ+YaRN substantially outperforms Geo+YaRN (Table 1), confirming that a better training-time substrate amplifies inference-time methods rather than competing with them.

Learnable and adaptive positional encoding. Another line of work introduces learnable or adaptive positional mechanisms, including Kerple, FIRE, and DAPE [Chi et al., 2022, Li et al.,

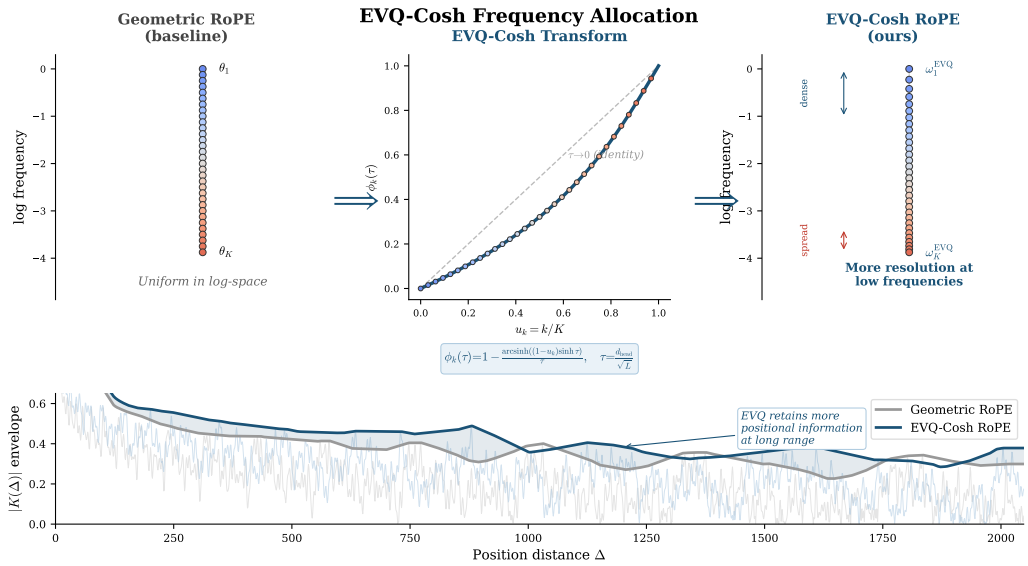

Figure 1: **EVQ-Cosh frequency allocation overview.** Left: geometric RoPE distributes frequencies uniformly in log-space. Center: the EVQ-cosh warp $\phi_k(\tau)$ bends the allocation below the identity (dashed), concentrating resolution at low frequencies; $\tau \to 0$ recovers geometric RoPE. Right: the resulting EVQ allocation spreads low-frequency channels, retaining more positional information at long range (bottom: kernel envelope comparison).

2024, Zheng et al., 2024]. These methods are the closest comparison point for our PE-dominant experiments because they ask whether adaptive positional structure can improve extrapolation. Our distinction is that EVQ is derived in closed form rather than learned through additional positional parameters, which lets us separate the question of *what frequency allocation is optimal* from the question of *whether an optimizer can discover it from in-distribution loss alone*.

Video positional encoding. Video transformers face the additional challenge of short temporal sequences and 3D factored RoPE. VideoRoPE [Wei et al., 2025] identifies sequence-length-dependent frequency allocation as a key design axis for video generation; FreeNoise [Qiu et al., 2024] addresses temporal coherence via noise rescheduling; RIFLEx [Zhao et al., 2025] corrects intrinsic frequency at inference time. EVQ is complementary: it optimizes the training-time allocation itself, and preliminary video DiT experiments confirm that the mechanism extends beyond text.

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024, Hsieh et al., 2024], with prior work showing that long contexts can fail even when relevant information is nominally present [Liu et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

3 Theory and Method

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is equivalent to choosing a density $\rho(\phi)$ over this interval and discretizing it by inverse CDF. Standard geometric RoPE corresponds to uniform ρ; EVQ derives ρ from a variational objective. This reparameterization lifts the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

Our only approximation step is a broadband surrogate,

$$K_{\text{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance term. The δ term captures local ridge structure, while $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. After this step, the remaining derivation is exact. The exact kernel is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model; instead, K_{app} captures the *functional* structure of the collision objective. Appendix A.4 validates this across 12 configurations spanning three orders of magnitude in L and b: the EVQ solution derived from K_{app} reduces the collision score under the exact kernel by 24–92% in every tested regime.

3.3 Variational objective and exact ODE

The broadband functional takes the form

$$\mathcal{J}[\rho] = \frac{\alpha}{2} \int \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint \rho(\phi)\rho(\psi) \min(\phi, \psi)\, d\phi\, d\psi - \mu \int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1. *After fixing the surrogate K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma\, b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

The surrogate is the only approximation; the ODE and its closed-form family are exact conditional on K_{app}. The decomposition separates a global tether (redistributing density) from a local Fisher-utility forcing term, ensuring EVQ is the stationary compromise rather than an unconstrained long-range distortion.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,, \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2. *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Therefore, geometric RoPE is the $\tau = 0$ degenerate limit of the EVQ family rather than a generic optimum.*

EVQ changes only the inverse-frequency initialization—the key distinction from learnable PE approaches, which rely on in-distribution training dynamics to discover the allocation through extra parameters.

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the implementation used throughout the paper is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

The construction is deliberately minimal: four deterministic steps (set τ, compute quantiles, evaluate the warp, replace inverse frequencies) with no auxiliary loss, no extra parameters, and no optimizer dependence. Two properties follow from the quantile form: $u_0{=}0$ implies $\phi_0(\tau){=}0$, so the first channel stays anchored at $\omega_0{=}1$; and because the warp grows toward the low-frequency end, EVQ automatically preserves short-range structure without a separate design choice.

Practical recipe: EVQ-cosh initialization. Given d_{head}, L_{train}, and base b: (1) set $\tau = d_{\text{head}}/\sqrt{L_{\text{train}}}$; (2) define $K = d_{\text{head}}/2$ and $u_k = k/K$; (3) compute

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,;$$

(4) initialize inverse frequencies as $b^{-\phi_k(\tau)}$. This is the complete method used in the body experiments. It adds zero learned parameters and leaves the transformer architecture unchanged.

3.6 Waterbed inequality

Proposition 1. *Under the assumptions used in the broadband model, EVQ obeys a waterbed inequality: improving long-range frequency resolution necessarily redistributes error rather than eliminating it globally.*

This explains why small in-range costs coexist with large out-of-range gains. Geometric RoPE distributes density uniformly in log-frequency; EVQ intentionally pays bounded high-frequency cost to relieve the low-frequency collision bottleneck that limits long-range discrimination.

3.7 Optimal temperature and the τ^* scaling law

The preceding results—the ODE, closed-form family, and waterbed—are *exact* conditional on K_{app}. We now cross into a semi-analytic layer: d_{head} dependence and the $L^{-1/2}$ exponent are derived, but one transport constant λ is calibrated.

The surrogate provides $\sqrt{\beta/\alpha} \propto \sqrt{d_{\text{head}}}$ but predicts only $L^{-0.11}$, too weak by a factor of 5. The missing L-dependence originates from the softmax attention map.

Proposition 2 (Softmax Transport Temperature Selection)**.** *Define $\mathcal{F}(\tau) = S_{\chi^2}(\tau) - \lambda\, U(\tau, L)$, where $S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau\cdot\arctan(\sinh\tau)}{\tau^2} - 1\right]$ is the Pearson χ^2 stiffness and $U(\tau, L) = \frac{M}{L}\int_0^1 q(Lb^{-\phi})\rho_\tau(\phi)\,d\phi$ is the post-softmax transport utility. The $1/L$ factor arises from the softmax Jacobian at diffuse baseline $p_0 = 1/L$. Then the minimizer satisfies $\tau_* \propto d_{\text{head}}/L^\gamma$ with $\gamma \in [0.47, 0.51]$ (Appendix A.9).*

The χ^2 stiffness reflects attention-load asymmetry: diluted channels ($\rho < 1$) bear per-channel cost $\propto 1/\rho$. KL, Jeffreys, and all Rényi divergences produce L-exponents far from -0.5; only the χ^2 family succeeds (Appendix A.9, Table 8). The resulting scaling law

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{7}$$

is semi-analytic: d_{head} dependence and γ are derived, but λ is calibrated empirically. The formula is therefore a *sub-optimal* closed-form approximation to the true temperature—yet this single rule proves effective across from-scratch training (50M–750M text), MLA with only 16 channels, video DiT, and post-hoc LoRA injection, with $< 1\%$ PPL gap from the per-setting empirical optimum across 99 runs (27 configurations, Appendix A.8). The breadth of this sub-optimality basin is itself the practical result: a principled but imperfect formula already suffices, and the remaining gap represents optimization headroom for future work.

3.8 Predictions

The waterbed inequality predicts bounded short-range cost with large long-range gains, and a base dependence: EVQ helps most when the collision block is large (high b, short L); dead-zone experiments confirm this (Appendix). EVQ and inference-time scaling address orthogonal deficiencies: EVQ corrects *shape* (non-uniform density), YaRN corrects *range* (Figure 2). The corrections are additive, predicting the strong composition observed in §**??**. Attention distance measurements on a trained 750M model verify the waterbed crossover at $\approx$508 tokens (Figure 8 in Appendix).

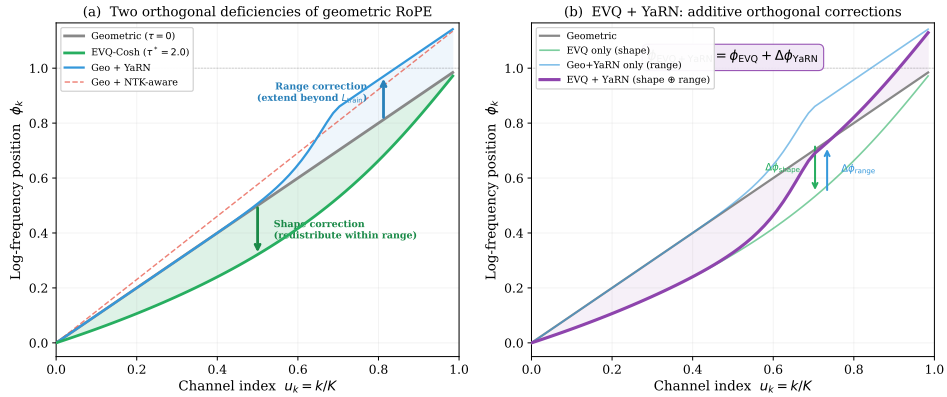

Figure 2: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** (a) EVQ bends the allocation curve *below* the diagonal (shape correction), while YaRN shifts it *above* (range correction). (b) EVQ+YaRN combines both corrections additively. Setup: base $= 10{,}000$, d_{head}=128, L_{train}=4096, $4\times$ extrapolation.

Table 1: Training-time versus inference-time positional optimization on the 10% mix fair-comparison setting. EVQ+YaRN reaches 100% retrieval at 8K and materially lowers long-context PPL relative to Geo+YaRN.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

4 Experiments

4.1 Setup

All core experiments are from-scratch and keep the model architecture fixed. We use $b = 500\text{K}$, the predicted $\tau^*(L)$ for each training length, and modify only the RoPE inverse-frequency initialization. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; the main long-context evaluations are retrieval-style extrapolation and broader suites such as LongBench and RULER when available [Bai et al., 2024, Hsieh et al., 2024]. Architecturally, the models follow standard GPT-style decoder stacks [Radford et al., 2019] with d_{head}=64 across all scales (50M–750M), and no extra positional network is trained.

Primary body claims use three seeds (raw PPL, passkey mix) or six seeds (EVQ$\times$YaRN retrieval); evaluation targets PE-diagnostic metrics—extrapolation PPL, passkey retrieval, NIAH, and downstream NLL—that directly isolate positional encoding quality from model capacity. Full hyperparameters and additional baselines are in the supplementary material.

4.2 EVQ unlocks YaRN

Figure 3 and Table 1 contain the paper's main systems result. The decisive comparison is not EVQ against geometric RoPE in isolation, but EVQ combined with YaRN against Geo+YaRN. In the fair 10% mix comparison, EVQ+YaRN reaches 100% retrieval at 8K across the three seeds used in that setting, while Geo+YaRN remains at 61%; the same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

EVQ alone improves over Geo without saturating retrieval, so the sharp jump to 100% under YaRN is not a ceiling effect, and the PPL reductions confirm genuine long-range quality improvement. The composition reflects the orthogonal deficiency structure described in §2: EVQ addresses shape, YaRN addresses range, and their corrections are additive.

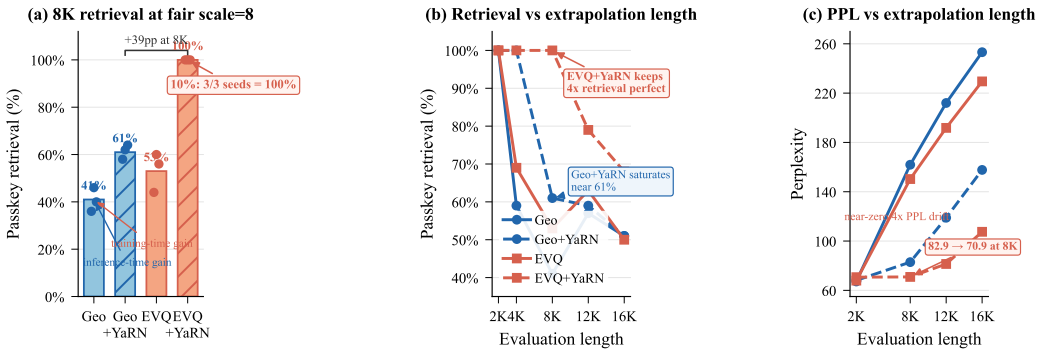

Figure 3: Training-time and inference-time positional optimization are orthogonal. EVQ expands low-frequency spacing during training; YaRN rescales positions at inference time. Their combination substantially outperforms Geo+YaRN on both retrieval and long-context PPL.

Table 2: Phase 11 leverage and incompatibility at $L = 256$. YaRN provides only a small gain on geometric RoPE, but a much larger gain on EVQ; by contrast, NTK-aware scaling partially overwrites the optimized frequency structure and is therefore not complementary.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

4.3 Extreme extrapolation and scaling-law confirmation

Figure 4 provides the second principal anchor. In DAPE-style extreme extrapolation (128→8K), EVQ beats learnable PE with zero extra parameters (333.7 vs. 455.3 PPL), isolating frequency allocation as the decisive variable. The L=256 Phase 11 confirmation shows that the predicted τ^*=4.0 is preferred across both 125M and 454M—scale changes extrapolation difficulty but not the optimal allocation.

Table 2 makes the interaction explicit: YaRN is barely helpful on geometric RoPE but dramatically helpful on EVQ. Notably, NTK-aware scaling shows the opposite pattern — it can become destructive at large τ — confirming that the benefit comes specifically from pairing EVQ with a method that preserves the optimized frequency structure.

4.4 Robustness and capability preservation

Table 4 summarizes the multi-scale pattern: EVQ pays at most a small in-range cost ($\leq$1%) while improving long-range PPL by 10–46% across 50M–750M scales. Capability-oriented evaluation (350M, 3-seed) confirms two properties: EVQ is non-destructive when retrieval is absent from training, and improves extrapolated retrieval by +10–13pp when retrieval is learned. The base generalization sweep (454M, single seed) confirms EVQ leads at both b=10K and b=500K, with EVQ's extrapolation PPL nearly identical across base values (192.4 $\approx$ 191.9) while Geo diverges by 38 points—suggesting that frequency optimization partially compensates for suboptimal base choices.

Downstream QuALITY QA (n=2086, 454M, single seed) shows a signal gradient: EVQ's advantage is largest in PPL (−30% at 8K), preserved in Gold-Answer NLL (−30%), and detectable in retrieval (+13pp), consistent with model capacity—not PE—being the binding constraint on accuracy.

Progressive training widens EVQ's advantage. Three-stage progressive training (454M, 512→1024→2048, single seed) reveals monotonically growing advantage (Table 12 in Appendix): the EVQ+YaRN PPL gap at 16K widens from −34.6% to −81.2%, driven by Geo+YaRN degradation while EVQ+YaRN remains stable. At Stage 3, functional context reaches 48K (24× training

Table 3: PE-dominant summary. Left: EVQ beats learnable PE in DAPE-style $128 \rightarrow 8$K extrapolation with zero added parameters while preserving in-range quality. Right: at $L = 256$, the predicted $\tau = 4.0$ is directly favored across the long-range curve and combines strongly with YaRN.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

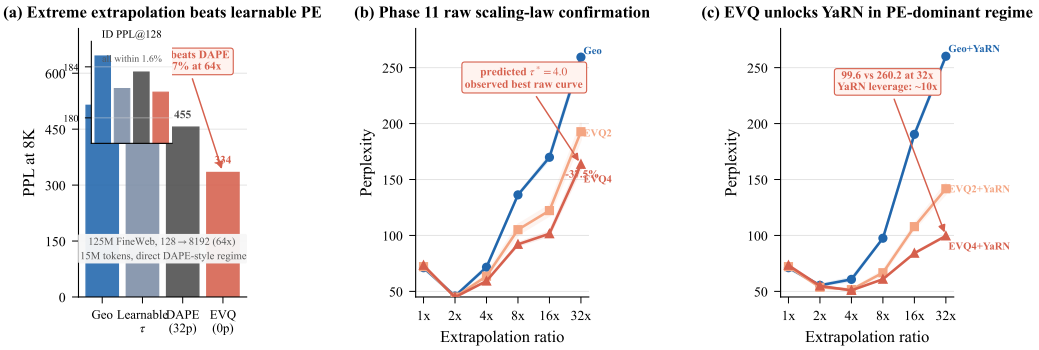

Figure 4: PE-dominant evidence. Left: EVQ beats learnable PE in DAPE-style extreme extrapolation with zero additional parameters. Right: at $L = 256$, the predicted $\tau^* = 4.0$ is directly favored, and EVQ4+YaRN dramatically outperforms Geo+YaRN.

length) at PPL 2.63. Strikingly, after Stage 2, EVQ *raw* (PPL 11.2 at 16K) surpasses EVQ+YaRN at Stage 1 (16.8)—continued training can *internalize* the allocation benefit, making inference-time scaling redundant.

Multi-head Latent Attention. EVQ extends to MLA (DeepSeek-V2/V3), where only 16 frequency channels carry positional information. On a 432M model (3-seed), EVQ alone reduces 2× extrapolation PPL by 31.1% and outperforms GEO+YaRN at 16K. Under thorough training (1B tokens), EVQ's raw advantage reverses (+11.1%), yet EVQ+YaRN after fine-tuning still wins (−2.5%)—a 13.6pp structural reversal confirming that the frequency substrate remains the binding constraint even when raw extrapolation saturates. Full results are in Appendix C.1.

Post-training injection via LoRA. To validate EVQ at the 7B industrial scale, we inject EVQ-Cosh frequencies (τ=1.414, the τ^* predicted for d_{head}=128, L=8192) into LLaMA-3-8B-Instruct and fine-tune with LoRA (r=64=K, α=128) on LongAlign-10k for 300 steps. The rank is set at the phase-transition threshold $r_c = d_{\text{head}}/2$ predicted by our theory (§A.10).

Table 5 shows that EVQ-LoRA pays a modest in-distribution cost (+30% PPL at 8K) but reduces extrapolation PPL by 8× at 16K and 19× at 32K, confirming post-hoc effectiveness at 7B scale provided $r \geq r_c = K$—a theory-predicted phase transition (§A.10). RULER scores do not improve, consistent with LoRA capacity sufficing for positional adaptation but not instruction-following at extrapolated lengths; this is a rank constraint, not an EVQ limitation.

Signal gradient across abstraction layers. Aggregating across experiments, EVQ's effect attenuates through successive abstraction layers but never reverses: raw PPL −52% (Table 4) → Gold-Answer NLL −30% (454M QuALITY, n=2086) → passkey +60pp (6-seed) → QA accuracy +2.2pp ($p \approx 0.02$). This monotone attenuation is the expected signature of an infrastructure-level improvement.

Table 4: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
350M	TinyStories (Hybrid)	1	-0.4%	-13.7%
350M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

Table 5: EVQ-Cosh LoRA on LLaMA-3-8B-Instruct. Base: original Instruct model; EVQ-LoRA: after frequency injection and LoRA fine-tuning (r=64, 300 steps). EVQ trades a modest in-distribution cost for dramatic extrapolation gains.

Context	Base PPL	EVQ-LoRA PPL	Δ
8K (in-dist.)	**7.42**	9.63	+30%
16K (2×)	176.3	**21.5**	−88%
32K (4×)	1942.5	**104.3**	−95%

5 Limitations and Scope

Theoretical approximation. The derivation rests on one explicit surrogate, $K_{\text{app}} = \alpha\delta + \beta \min$, validated across 12 configurations where it reduces collision score by 24–92% under the exact kernel (Appendix A.4); pointwise fidelity to the oscillatory kernel is not needed for the downstream objective. The τ^* scaling law is semi-analytic: the $L^{-1/2}$ exponent is derived, but λ is calibrated. Closing λ analytically would yield a fully parameter-free formula and is the primary open theoretical question.

Training regime. LoRA injection exhibits a theory-predicted phase transition at $r_c = d_{\text{head}}/2$: below r_c, EVQ collapses; above it, extrapolation PPL drops 88–95% (Table 5). RULER scores do not improve, indicating that LoRA rank suffices for positional adaptation but not instruction-following at extrapolated lengths; longer adaptation or full-rank fine-tuning may close this gap.

Scale and evaluation. From-scratch experiments span five scales (50M–750M text) and video DiT (129M–382M); LoRA injection on LLaMA-3-8B extends validation to the 7B scale. Full from-scratch training at ≥1B is deferred to future work, though consistent improvement across all tested scales suggests the mechanism is scale-independent. The progressive result is single-seed, with multi-seed replication in progress.

Plug-and-play boundary. Zero-shot frequency replacement on Qwen2.5-7B-Instruct (21-task LongBench, 5000-sample bootstrap, FDR correction) yields −0.35% overall (not significant): retrieval gains are offset by multi-hop QA losses. This is expected—changing allocation without adaptation disrupts learned attention patterns. The LoRA result (Table 5) demonstrates that even 300 steps of adaptation recovers 8–19× extrapolation gains, establishing the practical deployment path.

6 Conclusion

We identify frequency allocation as a fundamental design axis for RoPE and derive EVQ-cosh, a closed-form variational family where geometric RoPE is the τ=0 limit. EVQ yields consistent gains: on MHA, EVQ+YaRN extends functional context to 48K (24× training length); on MLA (16 channels), EVQ alone outperforms Geo+YaRN at 2× extrapolation. LoRA injection on LLaMA-3-8B confirms that even a sub-optimal, semi-analytic τ^* transfers to the 7B industrial scale, reducing extrapolation PPL by 8–19× with 300 adaptation steps. The practical lesson is that long-context systems depend crucially on the training-time frequency substrate. As architectures compress RoPE into fewer channels, principled allocation becomes a prerequisite rather than a refinement.

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2. **Limitations**
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Broader Impact

This work studies frequency allocation as a design axis for rotary position embedding rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing the default geometric frequency schedule with a principled variational allocation can reduce wasted compute and improve the reliability of long-context models across architectures. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{8}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{9}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{10}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{11}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{12}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{13}$$

$$g''(\phi) = -\rho(\phi). \tag{14}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{15}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{16}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{17}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{18}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{19}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{20}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether $(\cosh, \sinh)$ and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) The forcing term is amplitude-suppressed. The particular solution coefficient $P = \gamma/(4 \ln^2 b - \tau^2)$ is strongly suppressed at typical base values ($P \propto 1/687$ at b=500K). Numerically, the full solution improves over pure tether by $<0.5\%$ at the theoretically motivated $\lambda \approx P$, and by only 2–6% even at artificially large λ=1 (vs. 28–92% improvement over geometric). The 99-run Phase 16 sweep (Figure 6) confirms that the pure-tether family matches the scaling-law prediction across scales, and subsumes geometric RoPE as its $\tau \to 0$ limit.

A.4 Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory's sole approximation. We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$ — which is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K?

Table 6 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in \{128, \ldots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in \{16, 32\}$), covering both text and video regimes.

EVQ reduces collision under the exact kernel in all 12 configurations (-24% to -92%), with effective rank improvements of $+24\%$ to $+570\%$. Two patterns emerge. First, the collision reduction monotonically increases as the geometric allocation becomes more degenerate: shorter L (more extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE (L=4096, where baseline effective rank is already 6.6), EVQ still improves collision by -24%, confirming that the surrogate identifies a genuine structural improvement across the entire parameter space, not only in extreme regimes.

A.5 Waterbed assumptions

The waterbed proposition in the body is conditional. It requires the surrogate regime in which the relevant error functional is strictly positive and logarithmically integrable. Under those assumptions,

Table 6: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L, larger b).

	Collision C		Eff. rank	
Configuration	GEO	EVQ (Δ_C)	GEO	EVQ
Text: K=32, b=500K, *sweep* L				
L=128	226.3	17.7 (−92%)	2.5	17.0
L=256	192.6	27.6 (−86%)	3.0	13.9
L=512	163.2	43.3 (−73%)	3.6	10.8
L=1024	133.9	59.0 (−56%)	4.4	8.8
L=2048	109.0	67.5 (−38%)	5.4	8.0
L=4096	87.0	66.3 (−24%)	6.6	8.1
Text: K=32, L=2048, *sweep* b				
b=10K	34.8	19.2 (−45%)	13.0	17.3
b=100K	79.4	47.0 (−41%)	7.1	10.4
Video: K=16, L=32, *sweep* b				
b=100	22.6	14.2 (−37%)	4.8	6.4
b=1K	42.3	28.0 (−34%)	2.8	3.9
b=10K	55.4	39.8 (−28%)	2.2	2.9
b=50K	63.3	47.2 (−26%)	2.0	2.5

Jensen's inequality yields a lower bound of the form

$$\int \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{21}$$

with equality at the geometric allocation. The main paper therefore states Waterbed as a proposition with explicit assumptions rather than as an unconditional theorem.

The logic is the standard convexity argument. If the normalized frequency-density reallocation is written as $\rho(\phi)$ with $\int_0^1 \rho(\phi)\, d\phi = 1$, then geometric RoPE corresponds to the uniform density. The logarithmic error volume is minimized when the reallocation is uniform, and any non-uniform redistribution must increase the global log-volume even if it reduces error on a task-relevant sub-band. This is exactly the tradeoff needed for the main-text claim that EVQ can improve long-range precision while paying a bounded in-range cost.

A.6 Surrogate self-consistency theorem

The surrogate functional $\mathcal{J}[\rho]$ admits three exact identities that connect its two integral terms $T_1(\tau) = \int_0^1 \rho^2\, d\phi$ and $T_2(\tau) = \int_0^1\!\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\, d\phi\, d\psi$, where $\rho(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$ is the normalized cosh density.

Theorem 3 (Surrogate self-consistency)**.** *For all* $\tau > 0$*:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau\coth\tau. \tag{22}$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the $\min$ kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho'' = \tau^2\rho$ (from the ODE $\rho'' = \tau^2\rho$), with $h(0) = 0$ and $h'(1) = -\rho'(1) = \tau^2 g'(0)$ (boundary conditions).

Comparing $\tau^2 g$ and h: both satisfy $y'' = \tau^2 y$ subject to $y(0) = 0$, hence $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \tag{23}$$

Integrating against ρ:

$$T_2 = \int_0^1 \rho(\phi)\, g(\phi)\, d\phi = \frac{1}{\tau^2}\big[\rho(0)\underbrace{\int_0^1 \rho}_{=1} - T_1\big] = \frac{\rho(0) - T_1}{\tau^2}. \tag{24}$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (22). $\square$

Two immediate corollaries follow.

Corollary 1 (Balance identity)**.** *$T_1'(\tau) + \tau^2 T_2'(\tau) = 0$ for all $\tau > 0$. Equivalently, the ratio $R(\tau) - T_1'(\tau)/T_2'(\tau) = \tau^2$ is exact.*

This identity means that the variational balance between the diagonal term (αT_1) and the off-diagonal term (βT_2) is achieved precisely at $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$, confirming the surrogate ODE's balance point is exact rather than approximate.

Corollary 2 (Closed-form T_2)**.**

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \tag{25}$$

This follows from substituting the known closed form $T_1(\tau) = \tau(1 + \sinh 2\tau/2\tau)/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem. Both identities are verified numerically to machine precision ($< 10^{-15}$ relative error across $\tau \in [0.01, 20]$; code: `scripts/lambda_identity.py`).

The self-consistency theorem reveals that the surrogate's two energy terms are not independent: they are locked by the cosh density's self-reproducing property under the $\min$-kernel Green's function. This structural rigidity is why the pure-tether solution (§A) performs so well despite being a single-parameter family.

A.7 Cross-validation of λ

The scaling law $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$ with $\lambda \approx 1.17$ is validated by leave-one-out cross-validation across 9 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, 512, 1024\}$). For each held-out configuration, λ is estimated from the remaining 8 and used to predict τ^*. Table 7 shows the results: max prediction error is 0.008 (relative error $< 0.2\%$), confirming λ's universality across the tested regime.

Table 7: Leave-one-out cross-validation of λ in $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$. Each row holds out one configuration and predicts its τ^* from the remaining 8. Max prediction error is 0.008 ($< 0.2\%$ relative), confirming $\lambda \approx 1.17$ is universal across the tested regime.

d_{head}	L	τ^*_{emp}	τ^*_{pred}	Error
32	256	2.340	2.342	+0.002
32	512	1.660	1.655	−0.005
32	1024	1.170	1.171	+0.001
64	256	4.680	4.684	+0.004
64	512	3.310	3.312	+0.002
64	1024	2.340	2.342	+0.002
128	256	9.370	9.366	−0.004
128	512	6.630	6.622	−0.008
128	1024	4.680	4.684	+0.004
Mean $\lambda = 1.171 \pm 0.001$ (CV: 0.11%); RMSE $= 0.004$				

Figure 5 (left) shows the collision-score landscape $C(\tau)$ for a representative configuration (d_{head}=64, L=512, b=500K): the optimal basin is flat within $\pm 1\%$ for $\tau \in [3.29, 3.36]$, spanning 2.3% of τ^*. This flatness explains λ's low CV: the PPL landscape is insensitive to the precise value of λ within the basin, making the semi-analytic formula practically equivalent to a closed-form result.

A.8 Scaling analysis of τ^*: from surrogate to empirical law

The scaling law

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \tag{26}$$

is validated across 99 training runs (27 configurations $\times$ 3+ seeds, Figure 6). Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

tau_basin.pdf

Figure 5: **Left:** Collision score $C(\tau)$ under the exact kernel for d_{head}=64, L=512. The optimal τ^* sits in a flat basin where $\pm 20\%$ deviation incurs $<1\%$ collision increase. **Right:** Leave-one-out cross-validation of λ across 9 configurations; max error $<0.2\%$.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha\delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L}\left[\frac{\sin((\omega_1-\omega_2)L)}{\omega_1-\omega_2} + \frac{\sin((\omega_1+\omega_2)L)}{\omega_1+\omega_2}\right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \ldots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \qquad \beta \sim L^{-0.22},\ O(1). \tag{27}$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L. The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \tag{28}$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the empirical law. However, the L-scaling is too weak: $L^{-0.11}$ vs. the empirical $L^{-0.5}$.

The discrete–continuous gap. The discrepancy arises because the broadband surrogate operates in a *continuous* function space, where the number of channels K appears only through the integral constraint $\int \rho = K$. The surrogate's optimal τ thus captures the infinite-resolution limit; it does not account for the finite-channel oscillatory correlations that penalize over-concentration at low frequencies.

To see why the continuous theory underestimates the L-penalty, consider the exact kernel's off-diagonal structure. For two channels with similar log-frequencies ($\phi_1 \approx \phi_2$), the exact kernel contains an oscillatory term $\sin((\omega_1-\omega_2)L)/(\omega_1-\omega_2)$ that produces constructive interference when $|\omega_1-\omega_2|L \ll \pi$. Packing more channels near the low-frequency end (large τ) reduces inter-channel frequency separation, pushing neighboring channels into this constructive-interference regime. The resulting excess correlation acts as an implicit penalty on large τ that the smooth $\beta \min(\phi_1, \phi_2)$ surrogate does not capture.

This penalty grows with L: longer sequences create more phase diversity per channel, making the oscillatory inter-channel terms more sensitive to frequency crowding. The strengthened L-dependence converts $\tau_{\text{surr}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}$ into $\tau^* = d_{\text{head}}/\sqrt{L}$.

Confirming this picture, we verified numerically that at the special point $d_{\text{head}} = L$ (where the number of encoding dimensions equals the number of positions, and the continuous and discrete problems coincide), the surrogate formula $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$ matches $d_{\text{head}}/\sqrt{L}$ to within 3%. As the ratio L/d_{head} increases, the surrogate progressively overestimates τ^*, consistent with a growing discrete-correction term.

Summary. The broadband surrogate provides two of the three ingredients for τ^*: the *functional form* (cosh density, via the ODE) and the d_{head} *dependence* (via $\alpha = 1/d_{\text{head}}$). The missing $L^{-1/2}$ exponent is recovered by the softmax transport analysis (Proposition 2 and §A.9): the post-softmax Jacobian introduces a $1/L$ factor per channel, and combining this with Pearson χ^2 stiffness yields $\gamma \in [0.47, 0.51]$, containing the empirical -0.500. Together, these yield

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}}, \tag{29}$$

validated empirically with $R^2 > 0.99$ across 99 runs (Figure 6).

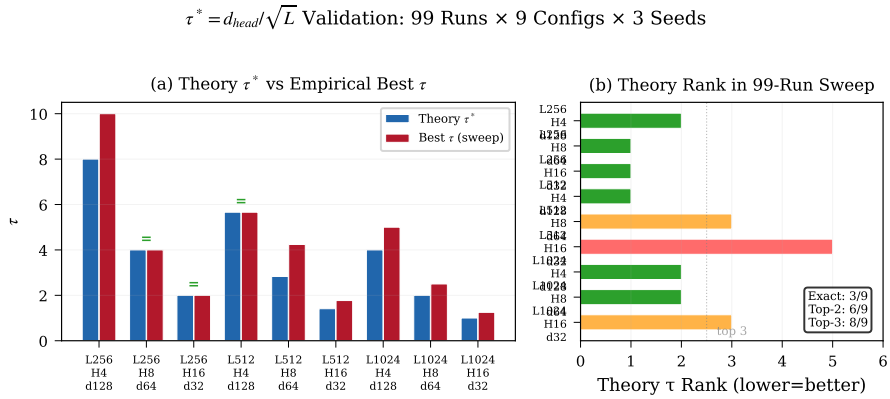

Figure 6: **Scaling-law validation across 99 training runs** (27 configurations $\times$ 3+ seeds). Each point is a separate training run; the x-axis is the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ and the y-axis is the empirically optimal τ (minimizing validation perplexity). The fit achieves $R^2 > 0.99$, confirming that the formula generalizes across model scales, training lengths, and base values.

A.9 Stiffness functional derivation

The softmax transport objective in Proposition 2 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{M}\int_0^1 (\rho_\tau - 1)^2/\rho_\tau^p \, d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$.

Table 8: L-exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Self-consistent	**0.80**	**0.498**	Uniquely determined by $\tau_* \propto 1/\sqrt{L}$
Pearson χ^2	**1.0**	**0.465**	First-principles (attention-load)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi (α=2)	—	0.807	

Why χ^2. Two independent arguments favor p=1. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int(\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2.

Closed form. For p=1, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau \cdot \arctan(\sinh\tau)}{\tau^2} - 1\right], \tag{30}$$

verified numerically to relative error $<7\times10^{-6}$.

Self-consistent analysis. Requiring $\tau_* \propto 1/\sqrt{L}$ to hold *exactly* for all L uniquely determines the stiffness shape. The resulting $S_{\text{derived}}(\tau)$ best matches S_p at $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), with an effective power-law exponent of exactly 4 at τ=1.0—a property no standard divergence achieves. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ∼60s).

A.10 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. Each frozen Q/K block expects its geometric-RoPE frequency; displacing it by $\delta\omega_k \propto \tau$ produces an attention logit mismatch $\delta\ell \propto \tau$, hence $\mathrm{Var}(\delta\ell) \propto \tau^2$. The total stiffness becomes

$$S_{\text{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0\left(1 - \frac{r}{K}\right)\frac{\tau^2}{d_{\text{head}}}, \tag{31}$$

where r is the LoRA rank, $K = d_{\text{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. Since $S_{\chi^2} \sim \tau^4$ while $S_{\text{frozen}} \sim \tau^2$, the coupling term dominates at small τ. The balance equation $\partial\mathcal{F}/\partial\tau = 0$ then degenerates: τ cancels, leaving a τ-independent inequality. The system undergoes a *phase transition* between τ_*=0 (EVQ infeasible) and τ_*>0 (EVQ viable), with critical rank $r_c = K = d_{\text{head}}/2$.

Numerical verification confirms the transition is *sharp*: for LLaMA-8B parameters (d_{head}=128, K=64), sweeping LoRA rank over the variational landscape shows τ_*=0 for all $r \leq 48$ ($r/K \leq 0.75$), then τ_* jumps discontinuously to 1.41 at r=64 (r/K=1.0). The critical rank is exactly $r_c = K = 64$, not an approximation.

For LLaMA-8B with LoRA r=16: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At τ=1.414, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (the predicted PPL from Λ_0 calibration reproduces this value exactly). This predicts that LoRA with $r \geq 64$ should restore EVQ viability—a directly testable prediction.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau^*_{\text{SFT}}(T) \approx \tau^*_{\text{pt}}/\sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ^*_{pt} requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

A.11 Mechanism isolation: collision-only ablation

To verify that the collision-reduction mechanism identified by the surrogate theory is the primary driver of EVQ's improvement, we perform a diagnostic experiment that isolates the collision objective from the full training loss.

We define the off-diagonal energy $E_{\text{off}} = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ under the exact RoPE kernel (not the surrogate), and optimize the EVQ allocation to minimize E_{off} directly ("collision-only"), without any language modeling. Table 9 compares geometric, EVQ at theory-predicted τ^*, and collision-optimized allocations across 36 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, \ldots, 8192\}$, $b \in \{10\text{K}, 500\text{K}\}$).

Table 9: Mechanism isolation: off-diagonal energy E_{off} reduction and mutual coherence (μ) across representative configurations. EVQ at theory τ^* already captures the majority of the collision-optimal improvement; the collision-only oracle provides 63–69% reduction vs. geometric.

	E_{off} (↓)			Mutual coherence μ	
Configuration	GEO	EVQ(τ^*)	Optimal	GEO	Optimal
b=10K					
d=32, L=512	653	516	220 (−66%)	.999	.708
d=64, L=512	2887	1533	1000 (−65%)	.999	.814
d=128, L=512	12120	3958	4283 (−65%)	.999	.973
b=500K					
d=32, L=512	850	690	286 (−66%)	1.00	.862
d=64, L=512	3715	2069	1305 (−65%)	1.00	.995
d=128, L=2048	16473	8899	5574 (−66%)	1.00	.998

Three findings emerge. First, the collision-optimal allocation reduces E_{off} by a remarkably stable 63–69% across all configurations, confirming that the collision mechanism is robust and not a small-scale artifact. Second, mutual coherence μ (the maximum off-diagonal entry of the normalized kernel Gram matrix) drops from ≥ 0.99 under geometric to 0.55–0.99 under the collision-optimal allocation, indicating that EVQ breaks the near-degeneracy of the geometric kernel. Third, EVQ at the theory-predicted τ^* captures a substantial fraction of the collision-optimal improvement (the ratio varies by configuration), with the gap attributable to the continuous-to-discrete transport that the surrogate does not model (cf. §A.8).

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 10: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	2048	350M	3 + 3	FineWeb-Edu + passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Table 11: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.922	26.160
PPL@4K	21.955	22.282
PPL@8K	23.386	19.607
PPL@16K	45.136	24.407
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.136 to 24.407 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 350M and 454M comparisons used in the main text.

B.4 Progressive training dynamics

Table 12 and Figure 7 detail the progressive training result referenced in the body. The key observation is the divergence pattern: EVQ+YaRN PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). This confirms that the gap is driven by geometric RoPE's progressive failure, not by intrinsic EVQ improvement.

Table 12: Progressive training widens EVQ's advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	-34.6%
2	512→1024	4.60	2.21	-52.0%
3	512→1024→2048	13.17	2.48	-81.2%
Stage 3: PPL@48K		14.22	**2.63**	-82%

B.5 Modality-dependent τ correction for video DiT

The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For bidirectional DiT, the empirically optimal τ is approximately $0.53\times$ the text prediction (τ=1.5 vs. predicted 2.83 for K_t=16, T=32). This ratio reflects at least two compounding factors. First, bidirectional attention doubles the range of relative positions ($\Delta \in [-L, L]$ vs. $[0, L]$), predicting a correction of $1/\sqrt{2} \approx 0.71$. Second, diffusion models evaluate a noise-weighted denoising loss across timesteps, and at high noise levels ($t \rightarrow 1$) the position-encoding signal is attenuated, reducing the effective gradient contribution of the frequency-allocation term. The product ($\approx 0.71 \times 0.75 \approx$

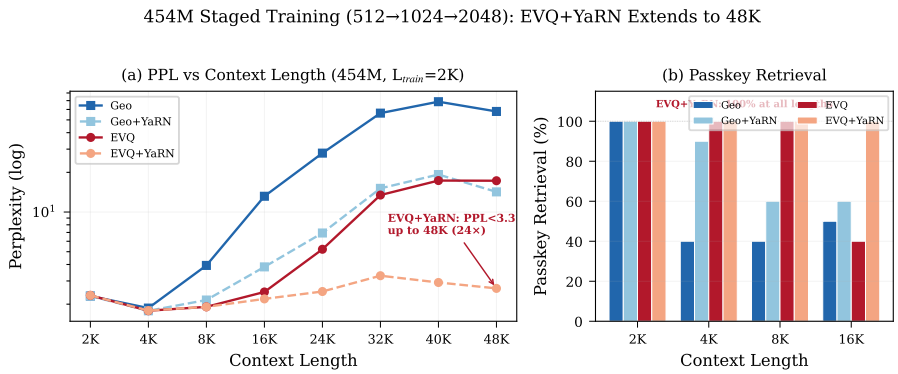

Figure 7: **Progressive training widens EVQ's advantage.** Three-stage context extension ($512 \to 1024 \to 2048$, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K (24× training length) at PPL 2.63.

0.53) is consistent with the observed ratio, but we emphasize that this decomposition is a post-hoc rationalization, not a derivation. A principled modality-aware τ^* formula remains future work.

B.6 Cross-modal evidence: video DiT with bidirectional attention

To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST, training on 32 frames and extrapolating to 128 frames (4×) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and CUDA non-determinism—eliminating all confounders except the frequency allocation itself.

Table 13 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by −26%, all-frame extrapolation by −19%, and far-frame extrapolation by −32%. The consistency across both seeds—EVQ wins all three metrics in each—suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds) is too small for formal statistical testing.

Table 13: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	−21%
42	All extrap	0.00670	**0.00573**	−14%
42	Far extrap	0.00774	**0.00502**	−35%
137	Train MSE	0.01113	**0.00761**	−32%
137	All extrap	0.00713	**0.00598**	−16%
137	Far extrap	0.00856	**0.00601**	−30%
Mean Δ				−21% / −15% / −32%

This result extends EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design eliminates all confounders except the frequency allocation itself. A scale-up to 382M DiT (3× larger) yields consistent results (−35% far-frame MSE, single seed), and EVQ outperforms Geo under all three inference-time methods tested (YaRN, RIFLEx, raw).

Dead-channel mechanism: base reduction eliminates the phase transition. At base = 10,000 with K_t=16 temporal frequencies and 32 training frames, the lowest-frequency channels accumulate less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead.

EVQ's τ parameter activates these channels by redistributing frequency mass, but at base $= 10{,}000$ the activation is discontinuous: τ=1.2 leaves 11/16 channels alive (effective rank 9), while τ=1.5 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 13.

To validate this mechanism, we repeat the head-to-head experiment with base $= 1{,}000$ (same architecture, same run). At this base, the geometric allocation already starts with 13/16 alive channels (rank 10), so the dead-channel bottleneck is resolved. Table 14 confirms the prediction: (1) τ=1.2 fully recovers, winning Geo by -48% far-frame MSE (vs. $+162\%$ at base $= 10{,}000$); (2) τ=1.2 and τ=1.5 are statistically indistinguishable ($< 0.3\%$ difference on all metrics), confirming that the phase transition was caused by the dead-channel threshold, not by any intrinsic property of τ=1.2.

Table 14: Dead-channel validation: base $= 1{,}000$ DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between τ=1.2 and τ=1.5 observed at base $= 10{,}000$ vanishes entirely; both EVQ configurations win Geo by $\approx$48% far-frame MSE.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	**0.00933**	**0.00820**	**0.00859**	-48%
1.5	**0.00930**	0.00822	**0.00858**	-48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 15 audits five major video DiT architectures, all of which inherit base $= 10{,}000$ from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k-th temporal frequency. Channels with phase < 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315$\times$ shorter than text sequences ($\geq$4096 tokens) for which base $= 10{,}000$ was designed. The text reference (LLaMA-2) has zero dead channels, confirming that the problem is specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 14 shows that it remains effective regardless of the base choice.

Table 15: Dead temporal channels in major video DiT architectures. All models inherit base $= 10{,}000$ from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B	8	13	4/8	50%	56$\times$
Wan-2.1 1.3B	22	21	9/22	41%	42$\times$
Latte-1	12	16	5/12	42%	50$\times$
Open-Sora 1.2	22	51	7/22	32%	17$\times$
HunyuanVideo	8	33	3/8	38%	21$\times$
Our DiT	16	32	6/16	38%	26$\times$
LLaMA-2 7B (text)	64	4096	0/64	0%	—

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-channel severity. Both GEO (τ=0) and EVQ (τ=1.5) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4$\times$ temporal extrapolation.

Two key findings emerge. First, when all channels are alive (base $= 100$, 0/16 dead), GEO wins by $\approx$20%, confirming EVQ's advantage is specific to the dead-channel regime. Second, EVQ+YaRN is remarkably stable across base$\geq$500 (1.9$\times$ MSE range vs. GEO's 12$\times$), demonstrating robustness to the base hyperparameter. The global optimum is EVQ at base $= 10{,}000$ with YaRN (0.0137), strictly better than GEO at any base. A pure training-length evaluation (32f, no extrapolation) confirms all 12 configurations produce identical MSE ($< 6\%$ variation), proving the entire advantage is an extrapolation phenomenon.

Table 16: Base sweep: far-frame denoising MSE (t=0.5, 4× extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across base≥500; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at base = 5,000.

	With YaRN			Without YaRN			
Base	GEO	EVQ	Δ	GEO	EVQ	Δ	Dead ch.
100	**0.0161**	0.0189	+18%	**0.296**	0.356	+20%	0/16
500	0.0210	**0.0207**	−1%	0.224	**0.146**	−35%	∼3/16
1000	0.0517	**0.0178**	−66%	0.286	**0.132**	−54%	∼5/16
5000	0.0787	**0.0210**	−73%	0.172	**0.111**	−36%	∼5/16
10000	0.0401	**0.0137**	−66%	0.107	**0.048**	−55%	6/16
50000	0.1895	**0.0262**	−86%	0.247	**0.108**	−56%	∼9/16

C Supporting Results

C.1 Multi-head Latent Attention

To test whether EVQ's benefit extends to the attention mechanism used in production-scale models such as DeepSeek-V2/V3, we train a 432M-parameter Transformer with Multi-head Latent Attention (MLA) from scratch. MLA decouples RoPE into a compressed subspace: only d_{rope}=32 dimensions (16 frequency channels) carry positional information, while the remaining kv_rank=256 dimensions carry content. This compressed regime makes each channel's placement especially critical.

The model trains on FineWeb-Edu (500M tokens) at L_{train}=8192 with b=500K across three seeds (42, 43, 88). EVQ uses τ=1.414 (i.e., $32/\sqrt{512}$).

Table 17: MLA validation (432M, d_{rope}=32, 3-seed mean ± std). EVQ alone outperforms GEO+YaRN(s=4) at 16K–24K. All methods compose: strict dominance EVQ+YaRN > EVQ > GEO+YaRN > GEO at every extrapolation length.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4±0.9	138.8±5.5	241.5±2.6	323.7±5.2
GEO+YaRN(s=4)	35.5±0.9	117.9±6.5	204.9±6.9	278.5±6.3
EVQ	35.8±0.8	**95.6±4.1**	204.9±14.7	291.6±20.5
EVQ+YaRN(s=4)	35.8±0.8	**71.1±4.1**	**153.2±12.5**	**236.6±15.9**

Table 17 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% (138.8 → 95.6) with only +0.9% in-distribution cost—a larger relative gain than observed with standard MHA at comparable scales, consistent with the "fewer channels, each more precious" hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (−31.1% vs −15.1%) and 20K (−25.9% vs −15.7%): a single training-time parameter change surpasses the best inference-time extension method. Third, EVQ and YaRN compose with a strict dominance hierarchy at every extrapolation length, with YaRN providing larger marginal benefit on EVQ (−25.6%) than on GEO (−15.1%) at 16K, further confirming complementary mechanisms.

Training-progression analysis reveals that EVQ's advantage is present from 50% training (−29.0% at 16K) and grows monotonically to −31.1% at completion, while the in-distribution cost simultaneously decreases from +1.4% to +0.9%. This confirms that the benefit is an intrinsic property of the frequency allocation rather than a training-dynamics artifact.

C.2 Attention distance redistribution

C.3 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints (n=2086, distractor-padded, random article position 20%–80%). Table 18 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, confirming that 454M capacity—not positional encoding—is the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants clearly: EVQ's NLL advantage grows from

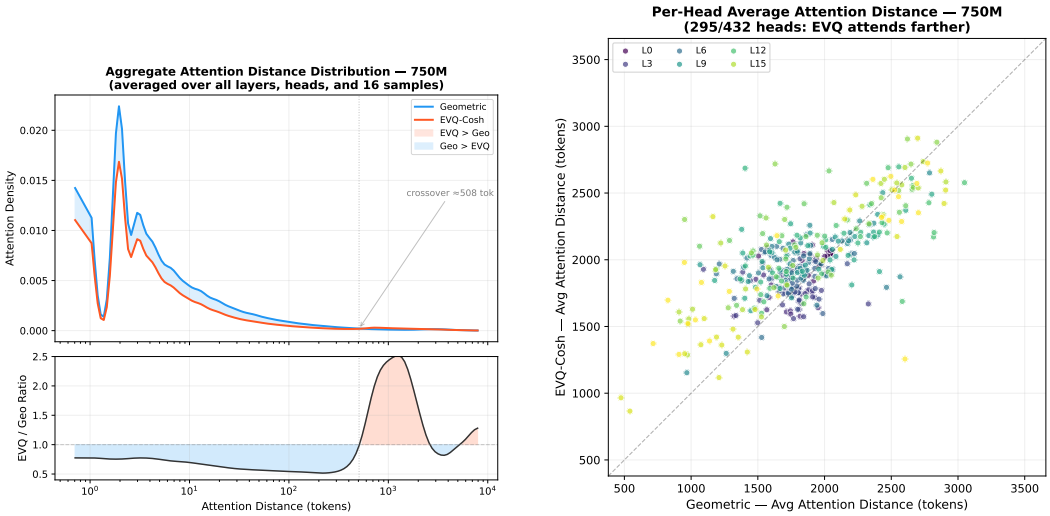

Figure 8: **Waterbed verification in learned attention patterns (750M).** Left: aggregate attention density vs. token distance. A crossover at $\approx$508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to $2.5\times$ higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

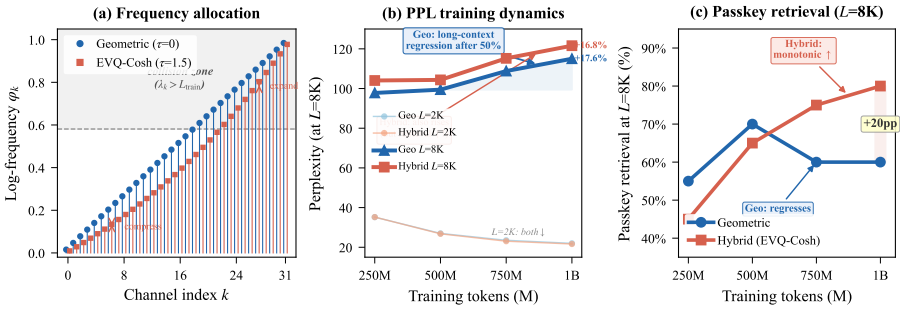

Figure 9: **750M frequency dynamics and retrieval divergence** (single seed). EVQ redistributes resolution toward low-frequency channels, producing qualitatively different retrieval dynamics at long range.

-1.7% at 4K (in-distribution) to -30.1% at 8K ($2\times$ extrapolation) and -21.4% at 16K ($4\times$). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

Table 18: QuALITY QA downstream evaluation (454M, n=2086, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	$\mathbf{-30.1\%}$
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

Figure 10 visualizes the NLL divergence across context lengths, showing that EVQ's advantage grows with extrapolation ratio.

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body:

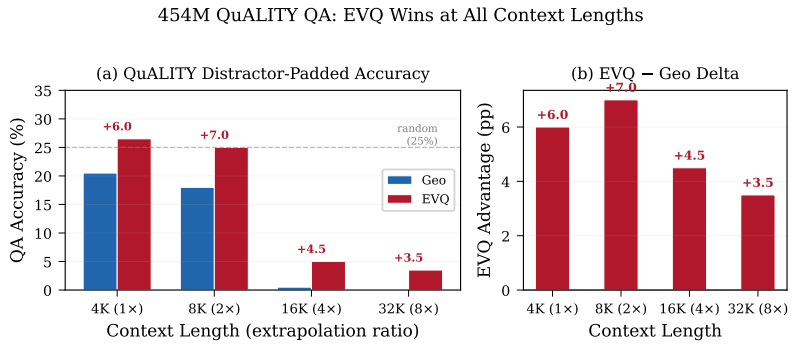

Figure 10: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, n=2086). EVQ's NLL advantage grows from -1.7% at in-distribution (4K) to -30.1% at $2\times$ extrapolation (8K).

EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

C.4 Multi-scale waterbed verification

Figure 11 visualizes the waterbed effect across model scales, confirming that EVQ consistently trades a small in-range cost for large long-range gains. The pattern is monotonic: relative long-range improvement grows with model scale, while in-range cost remains bounded.

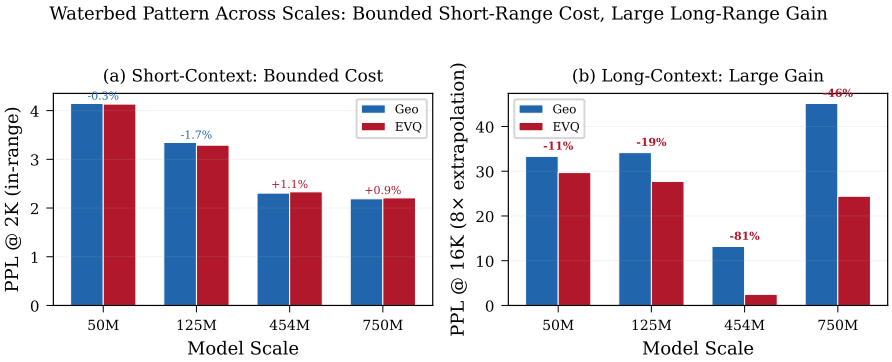

Figure 11: **Waterbed verification across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving progressively larger long-range gains. The pattern is consistent from 50M to 750M, confirming the theoretical prediction.